EUROPEAN SOCIETY OF CARDIOLOGY (ESC) & EUROPEAN SOCIETY OF HYPERTENSION (ESH)

Clinical Practice Guidelines for the Management of Arterial Hypertension

Joint European Guidelines (ESC / ESH) - Diagnostic Comparison with AHA/ACC

Official Reference: 2018/2024 ESC/ESH Guidelines (European Heart Journal 2018;39:3021-3104)

Official Web Portal: https://www.escardio.org/Guidelines/Clinical-Practice-Guidelines/Arterial-Hypertension-Ma

1. ESC Diagnostic Criteria & Definition

* Arterial hypertension is defined by ESC as sustained office SBP >=140 mmHg and/or office DBP >=90 mmHg.
* Confirmation requires repeated office measurements across 2-3 visits, or validated out-of-office monitorin

2. ESC/ESH Blood Pressure Classification Table

Category	Systolic (SYS)	Operator	Diastolic (DIA)
Optimal	Less than 120 mmHg	AND	Less than 80 mmHg
Normal	120 - 129 mmHg	and/or	80 - 84 mmHg
High-Normal	130 - 139 mmHg	and/or	85 - 89 mmHg
Grade 1 Hypertension	140 - 159 mmHg	and/or	90 - 99 mmHg
Grade 2 Hypertension	160 - 179 mmHg	and/or	100 - 109 mmHg
Grade 3 Hypertension	>=180 mmHg	and/or	>=110 mmHg
Isolated Systolic HTN	>=140 mmHg	AND	Less than 90 mmHg

3. Diagnostic Comparison: AHA/ACC vs. ESC/ESH (Key Point for Judges)

The 130-139 / 80-89 mmHg Reading (e.g. 135/85 mmHg):

* American Perspective (ACC/AHA 2017):
* Formally classified as 'Stage 1 Hypertension'.
* Rationale: Early proactive alert and prompt intervention to prevent clinical disease progression.
* European Perspective (ESC/ESH Guidelines):
* Formally classified as 'High-Normal Blood Pressure', NOT hypertension.
* Rationale: Avoiding overdiagnosis and overtreatment of healthy individuals; prevents medical labeling.

Direct Comparison Matrix:

Diagnostic Metric	AHA / ACC (American)	ESC / ESH (European)
Hypertension Diagnostic Cutoff	SBP >=130 or DBP >=80 mmHg	SBP >=140 or DBP >=90 mmHg
Classification of 135/85 mmHg	Stage 1 Hypertension	High-Normal Blood Pressure
Clinical Rationale	Early warning & active risk reduction	Avoiding overdiagnosis & overtreatment
Pharmacotherapy at 135/85 mmHg	Only if ASCVD risk >=10% or DM	No drugs; lifestyle only unless CVD proven

4. Target Blood Pressure Goals by Age Group

* Primary Target: Reduce blood pressure to <140/90 mmHg in all treated patients.
* Strict Target (<65 years): Aim for SBP 120-129 mmHg if well tolerated, and DBP 70-79 mmHg.
* Older Patients (>=65 years): Aim for SBP 130-139 mmHg to preserve vital organ and cerebral perfusion.

5. Out-of-Office Diagnostic Thresholds

* Home Blood Pressure (HBPM Mean): >=135/85 mmHg constitutes hypertension.
* Daytime Ambulatory BP (Daytime ABPM): >=135/85 mmHg constitutes hypertension.
* 24-Hour Ambulatory BP (24h ABPM): >=130/80 mmHg constitutes hypertension.
* Night-time Ambulatory BP (Night-time ABPM): >=120/70 mmHg constitutes hypertension (non-dipping risk).